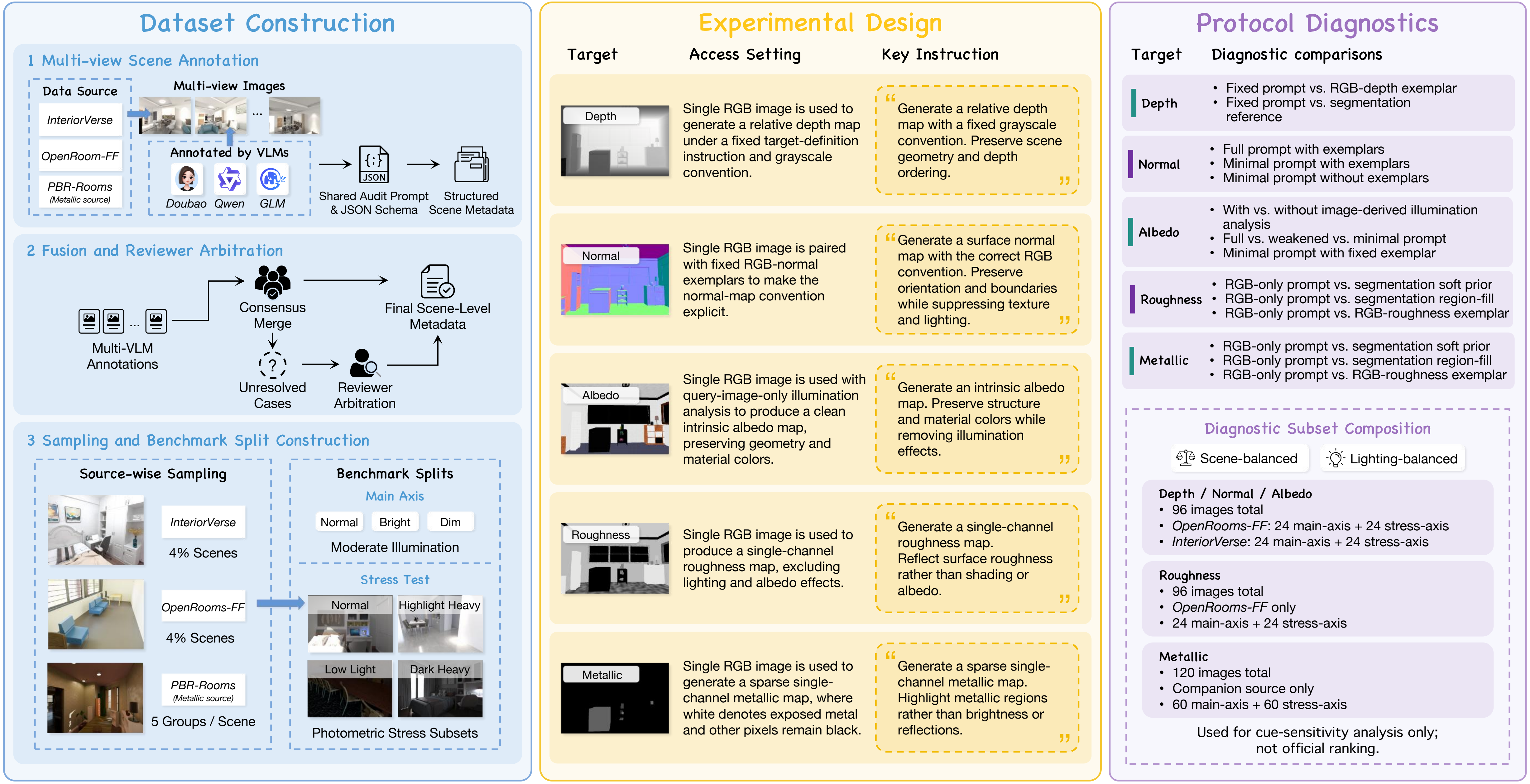

Dataset Construction
1 Multi-view Scene Annotation
Data Source
InteriorVerse
OpenRoom-FF
PBR-Rooms
(Metallic source)
Multi-view Images
Annotated by VLMs
Doubao
Qwen
GLM
JSON
Shared Audit Prompt & JSON Schema
Structured Scene Metadata
2 Fusion and Reviewer Arbitration
Multi-VLM Annotations
Consensus Merge
Final Scene-Level Metadata
Unresolved Cases
Reviewer Arbitration
3 Sampling and Benchmark Split Construction
Source-wise Sampling
InteriorVerse
4% Scenes
OpenRooms-FF
4% Scenes
PBR-Rooms
(Metallic source)
5 Groups / Scene
Benchmark Splits
Main Axis
Normal
Bright
Dim
Moderate Illumination
Stress Test
Normal
Highlight Heavy
Low Light
Dark Heavy
Photometric Stress Subsets
Experimental Design
Target
Access Setting
Key Instruction
Depth
Single RGB image is used to generate a relative depth map under a fixed target-definition instruction and grayscale convention.
Generate a relative depth map with a fixed grayscale convention. Preserve scene geometry and depth ordering.
Normal
Single RGB image is paired with fixed RGB-normal exemplars to make the normal-map convention explicit.
Generate a surface normal map with the correct RGB convention. Preserve orientation and boundaries while suppressing texture and lighting.
Albedo
Single RGB image is used with query-image-only illumination analysis to produce a clean intrinsic albedo map, preserving geometry and material colors.
Generate an intrinsic albedo map. Preserve structure and material colors while removing illumination effects.
Roughness
Single RGB image is used to produce a single-channel roughness map, excluding lighting and albedo effects.
Generate a single-channel roughness map. Reflect surface roughness rather than shading or albedo.
Metallic
Single RGB image is used to generate a sparse single-channel metallic map, where white denotes exposed metal and other pixels remain black.
Generate a sparse single-channel metallic map. Highlight metallic regions rather than brightness or reflections.
Protocol Diagnostics
Target
Diagnostic comparisons
Depth
• Fixed prompt vs. RGB-depth exemplar
• Fixed prompt vs. segmentation reference
Normal
• Full prompt with exemplars
• Minimal prompt with exemplars
• Minimal prompt without exemplars
Albedo
• With vs. without image-derived illumination analysis
• Full vs. weakened vs. minimal prompt
• Minimal prompt with fixed exemplar
Roughness
• RGB-only prompt vs. segmentation soft prior
• RGB-only prompt vs. segmentation region-fill
• RGB-only prompt vs. RGB-roughness exemplar
Metallic
• RGB-only prompt vs. segmentation soft prior
• RGB-only prompt vs. segmentation region-fill
• RGB-only prompt vs. RGB-roughness exemplar
Diagnostic Subset Composition
Scene-balanced
Lighting-balanced
Depth / Normal / Albedo
• 96 images total
• OpenRooms-FF: 24 main-axis + 24 stress-axis
• InteriorVerse: 24 main-axis + 24 stress-axis
Roughness
• 96 images total
• OpenRooms-FF only
• 24 main-axis + 24 stress-axis
Metallic
• 120 images total
• Companion source only
• 60 main-axis + 60 stress-axis
Used for cue-sensitivity analysis only; not official ranking.